Assignment 1

Category A - glass.csv

1、读取数据（安装库）

```
# First method import - 直接导入
glass <- read.csv("D:/USYD/5048/A1/glass.csv")

# Or using readr - 下载readr再导入
install.packages("readr")

library(readr)
glass <- read_csv("D:/USYD/5048/A1/glass.csv")

# Or import from RStudio, click File and click Import Dataset, select From
Text(readr)
# 与第二种类似，也会先下载readr
```

2、预处理数据

```
# Extract numeric variables (exclude last column Type) - 提取数值型变量，例如Type
# First
numeric_data <- glass[, c("RI", "Na", "Mg", "Al", "Si", "K", "Ca", "Ba", "Fe")]

# Or second
numeric_data <- glass[, c(1:9)]

# or Third
numeric_data <- glass[, ncol(glass)]
```

3、计算MDS

```
# Standardized data - 标准化数据
scaled_data <- scale(numeric_data)

# Calculate pairwise distances between data - 计算数据之间的成对距离
glass_scaled <- dist(scaled_data)

# Perform MDS, downscaling to 2 dimensions - 执行MDS，降维到2维
glass_dist <- cmdscale(glass_scaled, k=2)

# Add Type column - 把Type添加会MDS，方便上色
mds_glass <- data.frame(MDS1 = glass_dist[, 1],
                        MDS2 = glass_dist[, 2],
                        Type = as.factor(glass$Type))
```

4、绘制散点图

```
plot(x = mds_glass$MDS1, y = mds_glass$MDS2, col=mds_df$Type, type="p",
+      main = "MDS", xlab = "MDS 1", ylab = "MDS 2")
```